

Sheaves

All the section $f : X \rightarrow Y$, $\mathcal{F} \in \text{Sh}(X), \mathcal{G} \in \text{Sh}(Y)$

Def. *Sheaf*

- *locality*: $s|_{U_i} = t|_{U_i} \Rightarrow s = t$
- *gluing*: $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j} \Rightarrow \exists s \in \mathcal{F}(U), s|_{U_i} = s_i$

Def (Direct image). $f_*\mathcal{F}(V) := \mathcal{F}(f^{-1}(V))$

Def (Inverse image). $f^{-1}\mathcal{G}(U) := \lim_{\rightarrow f(U) \subseteq V} \mathcal{G}(V)$ (*presheaf*)

Def (Stalk). $\mathcal{F}_x := \lim_{\rightarrow x \in U} \mathcal{F}(U)$

Prop. $x \in U \Rightarrow \mathcal{F}(U) \rightarrow \mathcal{F}_x$

Def. $\mathcal{F}^+(U) := \{s : U \rightarrow \bigsqcup_{x \in U} \mathcal{F}_x : s(x) \in \mathcal{F}_x, \exists V \subseteq U \text{ st } t_y = s(y)\}$

Prop. $\mathcal{F} \rightarrow \mathcal{F}^+$ st $s \rightarrow (x \mapsto s_x), F_x^+ \cong \mathcal{F}_x$

Prop. $\varphi : \mathcal{F} \rightarrow \mathcal{F}'$. Only $\ker(\varphi)$ sheaf.

Prop. $0 \rightarrow \mathcal{F} \rightarrow \mathcal{G} \rightarrow \mathcal{H} \rightarrow 0$ ses $\Leftrightarrow 0 \rightarrow \mathcal{F}_x \rightarrow \mathcal{G}_x \rightarrow \mathcal{H}_x \rightarrow 0$ ses $\forall x \in X$

Prop. $(f^{-1}\mathcal{G})_x \cong \mathcal{G}_{f(x)}$

Prop. $(f_*\mathcal{F})_y \cong \lim_{\rightarrow y \in V} \mathcal{F}(f^{-1}(V))$

Prop. f^{-1} exact, f_* left exact.

Prop. Global section functor $\Gamma(X, -) : \mathcal{F} \mapsto \mathcal{F}(X)$ left exact.

Def. • (Ringed space) $(X, \mathcal{O}_X)$. X top. space, $\mathcal{O}_X$ sheaf of rings on X .

- (+ locally) $\mathcal{O}_{X,x}$ local ring $\forall x \in X$.
- (Affine scheme) $X \cong \text{Spec}(A)$.
- (Scheme) $X = \bigcup U_i, U_i \cong \text{Spec}(A_i)$.

Def (Morphism loc. ringed spaces). $(f, f^\#), f : X \rightarrow Y, f^\# : \mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$, local hom.

Prop. $\text{AffSch} \leftrightarrow (\text{Ring})^{\text{op}}, X \mapsto \mathcal{O}_X, (f, f^\#) \mapsto f^\#(Y)$

Prop. Adjunction $f^{-1} \dashv f_*$. $\text{Hom}_{\text{Sh}(X)}(f^{-1}\mathcal{G}, \mathcal{F}) \cong \text{Hom}_{\text{Sh}(Y)}(\mathcal{G}, f_*\mathcal{F})$

Def. $\text{supp}(\mathcal{F}) := \{x \in X : \mathcal{F}_x \neq 0\}, \text{supp}(s) := \{x \in U : s_x \neq 0\}$

Prop. $\text{Spec}(A \times B) \cong \text{Spec}(A) \sqcup \text{Spec}(B)$

Schemes and properties

We state (U_i) open cover, (W_i) open affine cover and U open subset, Z closed subset.

Also, $x \in X, y \in Y$ arbitrary point, $\eta \in X$ the generic point, $f : X \rightarrow Y$.

Def. • (Disconnected) $X = U_1 \cup U_2, U_1 \cap U_2 = \emptyset$. e.g. $\text{Spec}(k \times k)$.

- (Reducible) $X = Z_1 \cup Z_2, Z_1, Z_2$. e.g. $\text{Spec}(k[x, y]/(xy))$.

- (Quasi-compact) $X = \bigcup U_i$, finite subcover. e.g. not $\text{Spec}(k[x_1, x_2, \dots]) - \{0\}$.
- (Quasi-separated) U_i, U_j quasi-compact $\Rightarrow U_i \cap U_j$ quasi-compact. e.g. $U_i = \mathbb{A}_k^\infty$ glued along $\mathbb{A}_k^\infty - \{0\} \Rightarrow U_1 \cap U_2 = \mathbb{A} - \{0\}$.

Prop. irreducible $\Rightarrow$ connected

Prop. X has a generic point $\Leftrightarrow X$ irreducible.

Prop. Affine $\Rightarrow$ quasi-compact + quasi-separated

Def (Open immersion). $f : (U, \mathcal{O}_U) \rightarrow (X, \mathcal{O}_X), f^\#$ iso

Def (Closed immersion). $\iota : Z \hookrightarrow X, \begin{cases} \iota \text{ homeo. onto closed subset} \\ \iota^\# : \mathcal{O}_Y \twoheadrightarrow \iota_*\mathcal{O}_X \end{cases}$

Prop. Closed subschemes of aff. are aff.

Prop. $D(f) \rightarrow X$ open immersion, $V(I) \rightarrow X$ closed immersion.

Prop. $\text{Spec } B \rightarrow \text{Spec } A$ closed imm. $\Leftrightarrow \varphi : A \rightarrow B$ surjective.

Prop. $Z \subseteq \text{Spec}(A) \Leftrightarrow Z = \text{Spec}(A/I)$.

Def. • (affine-local) $(a_1, \dots, a_n) = A, A \in \mathcal{P} \Leftrightarrow A_{a_i} \in \mathcal{P}$
• (stalk-local) $A \in \mathcal{P} \Leftrightarrow A_{\mathfrak{p}} \in \mathcal{P}$

Prop. Stalk-local $\Rightarrow$ affine-local ($A_{\mathfrak{p}} \cong (A_{\mathfrak{a}})_{\mathfrak{p}A_{\mathfrak{a}}}$)

Def. • X Noetherian $\Leftrightarrow$ quasi-compact + A_i Noeth. (affine-local) (e.g. not stalk-local $A = \Pi_\infty \mathbb{F}_2$)

- X integral $\Leftrightarrow \mathcal{O}_X(U)$ integral $\Leftrightarrow X$ connected + $\mathcal{O}_{X,x}$ integral $\Leftrightarrow X$ reduced + irred.
- X reduced $\Leftrightarrow \mathcal{O}_X(U)$ reduced (stalk-local)
- X normal $\Leftrightarrow A_{\mathfrak{p}}$ integral and integrally closed (stalk-local)

Prop. $\text{Spec}(A)$ irred. $\Leftrightarrow A/\sqrt{(0)}$ integral.

Def (Residue / Function field). $k(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x, k(X) = \mathcal{O}_{X,\eta}$

Prop. $\mathcal{O}_X(U) = \bigcap_{x \in U} \mathcal{O}_{X,x}$

Def (Ideal sheaf). $\mathcal{I} : U \mapsto \mathcal{I}(U) \subseteq \mathcal{O}_X(U)$ ideal.

Def. • (fiber) $X_y := f^{-1}(y) = X \times_Y \text{Spec } k(y)$

- (geometric fiber) $X_{\bar{y}} := X \times_Y \text{Spec } \overline{k(y)}$
- (diagonal morphism) $\Delta_{X/Y} := X \rightarrow X \times_Y X, p_1 \circ \Delta = p_2 \circ \Delta = \text{id}_X$
- (graph) $\Gamma_f := (\text{id}, f) : X \rightarrow X \times_Y Y$

$$\begin{array}{ccccc} X \times_S Y & \longrightarrow & X \times_T Y & & X \times_Y Y' & \xrightarrow{f_{Y'}} & Y' & & X & \xrightarrow{j_{\Gamma_f}} & X \times_S Y \\ \downarrow & \lrcorner & \downarrow & & p_1 \downarrow & \lrcorner & \downarrow g & & f \downarrow & & \downarrow f_Y \\ S & \longrightarrow & S \times_T S & & X & \xrightarrow{f} & Y & & Y & \xrightarrow{\Delta_{Y/S}} & Y \times_S Y \end{array}$$

Prop. X, Y, S affine $\Rightarrow X \times_S Y$ affine

Def. $\mathcal{P}$ satisfies

- (COMP) $f, g \in \mathcal{P} \Rightarrow g \circ f \in \mathcal{P}$
- (CANC) $g \circ f \in \mathcal{P} \Rightarrow f \in \mathcal{P}$
- (BC) $f \in \mathcal{P} \Rightarrow f_{Y'} \in \mathcal{P}$
- (LOCT) $f \in \mathcal{P} \Leftrightarrow f_{V_i} \in \mathcal{P} \forall i (\bigcup V_i = Y)$
- (LOCS) $f \in \mathcal{P} \Leftrightarrow f|_{U_i} \in \mathcal{P} \forall i (\bigcup U_i = X)$

Def. $f \in \Delta_{\mathcal{P}} \Leftrightarrow \Delta_{X/Y} \in \mathcal{P}$

Prop. • $\mathcal{P}$ (BC) $\Rightarrow \Delta_{\mathcal{P}}$ (BC)

- $\mathcal{P}$ (BC, COMP) $\Rightarrow \Delta_{\mathcal{P}}$ (COMP)
- $\mathcal{P}$ (LOCT) $\Rightarrow \Delta_{\mathcal{P}}$ (LOCT)
- $\mathcal{P}$ (BC, COMP), $f, g \in \mathcal{P} \Rightarrow f \in \mathcal{P}$
- $\mathcal{P}$ (BC, COMP), $g \circ f, \Delta_{Y/Z} \in \mathcal{P} \Rightarrow f \in \mathcal{P}$

Def. • (quasi-compact) V q-c $\Rightarrow f^{-1}(V)$ q-c ($\Leftrightarrow V$ open affine)

- (quasi-separated) $\Delta_{X/Y}$ quasi-compact
- (finite fibers) $f^{-1}(y)$ finite set

Prop. all (COMP, LOCT) +

- injective (CANC)
- surjective (BC)
- open (LOCS)
- closed
- quasi-compact (BC)
- quasi-separated (BC, CANC)
- finite fibers (CANC)

Def. • (immersion) $f = g \circ h$, g, h open and closed imm. resp.

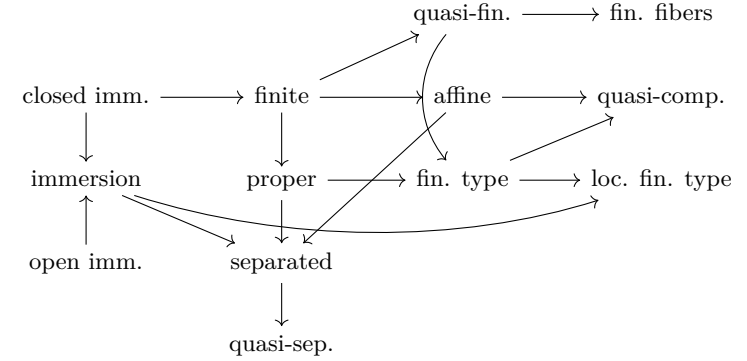
- (loc. fin. type) $V = \text{Spec } B \Rightarrow f^{-1}(V) = \bigcup \text{Spec } A_i, A_i \text{ f.g. } B\text{-alg.}$
- (fin. type) loc. fin. type + quasi-compact
- (affine) V affine $\Rightarrow f^{-1}(V)$ affine
- (quasi-finite) fin. type + finite fibers
- (finite) $V = \text{Spec } B \Rightarrow f^{-1}(V) = \text{Spec } A, A \text{ f.g. } B\text{-mod.}$
- (separated) $\Delta_{X/Y} : X \rightarrow X \times_Y X$ closed imm.
- (proper) separated + fin. type + univ. closed

Prop. all (COMP, BC, LOCT) +

- immersion (closed/open):
- loc. fin. type: (CANC, LOCS) (affine-local)

- fin. type:
- affine: (affine-local)
- quasi-finite:
- finite: (affine-local)
- separated: (CANC)
- proper:

Prop. Finite $\Rightarrow$ quasi-finite + proper



$$\boxed{M \otimes_R \frac{R}{I} \cong \frac{M}{IM}} \quad \boxed{B \otimes_R \frac{R[x]}{I} \cong \frac{B[x]}{IB[x]}} \quad \boxed{M \otimes_R S^{-1}R \cong S^{-1}M} \quad \boxed{S^{-1} \left(\frac{M}{N} \right) \cong \frac{S^{-1}M}{S^{-1}N}}$$

Proj

$A = \bigoplus_{d \geq 0} A_d$ graded ring, A_0 subring, $A_+ = \bigoplus_{d > 0} A_d$ irrelevant ideal.

Def (Homo. ideal). $f \in I \Rightarrow f_{d_1}, \dots, f_{d_r} \in I \Leftrightarrow I = \bigoplus_{d \geq 0} (I \cap A_d) \Leftrightarrow I = (f_1, \dots, f_r), f_i$ homo.

Def (Homogenization). $I^h = \bigoplus_{d \geq 0} (I \cap A_d)$

Prop. I homo $\Rightarrow \sqrt{I}$ homo.

Def. $M_{(a)} := (M_a)_0, \quad M_{(\mathfrak{p})} := (M_{\mathfrak{p}})_0$. e.g. $A = k[x_0, \dots, x_n], A_{(x_i)} = k[\frac{x_0}{x_i}, \dots, \frac{x_n}{x_i}]$

Prop. $D_+(f), f \in A_+$ homo, basis of topology.

$$\boxed{\text{Proj } A := \{\mathfrak{p} \text{ homo}, A_+ \not\subseteq \mathfrak{p}\}} \quad \boxed{V_+(I) := \{\mathfrak{p} \in \text{Proj } A : I \subseteq \mathfrak{p}\}} \quad \boxed{\overline{f(V(I))} = V(\varphi^{-1}(I))}$$

$$\boxed{D_+(f) := \{\mathfrak{p} \in \text{Proj } A : f \notin \mathfrak{p}\} \cong \text{Spec}(A_{(f)})} \quad \boxed{V_+(I \cap J) = V_+(I) \cup V_+(J)}$$

$$\boxed{V_+(\sum I_i) = \bigcap V_+(I_i)} \quad \boxed{V_+(A_+) = V_+(A) = \emptyset} \quad \boxed{V_+((0)) = \text{Proj } A}$$

Prop. $\boxed{\mathcal{O}_X(m)}$ $U_i = \text{Spec } A_i, \quad A_i = k[\frac{x_0}{x_i}, \dots, \frac{x_n}{x_i}], \quad M_i = A_i x_i^m$
 $(M_i)_{(\frac{x_j}{x_i})} \cong (M_j)_{(\frac{x_i}{x_j})}, \quad \Gamma(\mathbb{P}^n, \mathcal{O}_X(m)) = \text{free mod. hom. poly. deg. } m$

Def. $\mathbb{P}_X^n := X \times_{\text{Spec } \mathbb{Z}} \mathbb{P}_{\mathbb{Z}}^n, \quad \mathbb{P}_{A_0}^n = \bigcup_{i=0}^n \left(V(\sum_{j=0}^{i-1} (x_j)) \setminus V(\sum_{j=0}^i (x_j)) \right)$

Def. $\text{Spec } A$ affine cone over $\text{Proj } A, \text{Spec } A \setminus V(A_+) \rightarrow \text{Proj } A$

Def. Segre embedding $\sigma : \mathbb{P}^n \times \mathbb{P}^m \rightarrow \mathbb{P}^{(n+1)(m+1)-1}, \quad ([x_i], [y_j]) \mapsto [x_i x_j]$

Prop. A f.g. in deg. 1 $\Rightarrow \text{Proj } A \rightarrow \mathbb{P}_{A_0}^n$ closed immersion. i.e. $A_0[x_0, \dots, x_n] \twoheadrightarrow A \Rightarrow \text{Proj } A \rightarrow \mathbb{P}_{A_0}^n$

Prop. A f.g. in deg. 1 $\Rightarrow \pi_A \text{Proj } A \rightarrow \text{Spec } A_0$ proper.

Def. Veronese embedding $v_d : \mathbb{P}^n \rightarrow \mathbb{P}^N, \quad [x_i] \mapsto [x_i^I]$ monomials deg. $d. N = \binom{n+d}{n} - 1$

Prop. $\text{Proj}(S \times_A T) \cong \text{Proj}(S) \times_A \text{Proj}(T)$ and $\mathcal{O}(1) \cong p_1^*(\mathcal{O}_X(1) \otimes p_2^*(\mathcal{O}_Y(1)))$

Quasi-coherent sheaves

$\mathcal{F}, \mathcal{G} \in \text{Mod}(X, \mathcal{O}_X).$

Prop. $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}$ presheaf

Def (Pushforward). $f_* \mathcal{F}(V) = \mathcal{F}(f^{-1}(V))$

Def (Pullback). $f^* \mathcal{G} = f^{-1} \mathcal{G} \otimes_{f^{-1} \mathcal{O}_Y} \mathcal{O}_X$

Def. $\bullet$ (free) $\mathcal{F} \cong \mathcal{O}_X^{\oplus I}$
 $\bullet$ (loc. free) $\mathcal{F}|_{U_i}$ free
 $\bullet$ (rank r) $\mathcal{F}|_{U_i} \cong \mathcal{O}_{U_i}^{\oplus r}$
 $\bullet$ (invertible) rank 1 loc. free
 $\bullet$ (g.g.) $\exists \mathcal{O}_X^{\oplus n} \twoheadrightarrow \mathcal{F}$

Def. $\tilde{M}$ such that $\tilde{M}(D(f)) = M_f$

Def. $\bullet$ (quasi-coherent) $\forall U_i \cong \text{Spec } A_i, \mathcal{F}|_{U_i} \cong \tilde{M}_i, M_i$ A_i -module.
 $\bullet$ (coherent) $\varphi : \mathcal{O}_{U_i}^n \rightarrow \mathcal{F}|_{U_i}, \ker \varphi$ finite type.

Prop. $\mathcal{F}$ quasi-coherent $\Leftrightarrow \mathcal{O}|_{U_i}^{\oplus I_i} \rightarrow \mathcal{O}_{U_i}^{\oplus J_i} \twoheadrightarrow \mathcal{F}|_{U_i} \rightarrow 0$

Prop. $\text{Hom}_A(N, M) \cong \text{Hom}_{\mathcal{O}_X}(\tilde{N}, \tilde{M})$

Prop. $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$ ses $\Leftrightarrow 0 \rightarrow \tilde{M}_1 \rightarrow \tilde{M}_2 \rightarrow \tilde{M}_3 \rightarrow 0$ ses

Prop. $\widetilde{\ker(\varphi)} = \ker(\tilde{\varphi}), \widetilde{\text{Im}(\varphi)} = \text{Im}(\tilde{\varphi}), \widetilde{\text{Coker}(\varphi)} = \text{Coker}(\tilde{\varphi})$

Prop. $\bullet A - \text{Mod} \leftrightarrow \text{QCoh}(\text{Spec } A, \mathcal{O}_{\text{Spec } A})$
 $\bullet$ graded $A - \text{Mod} \leftrightarrow \text{QCoh}(\text{Proj } A, \mathcal{O}_{\text{Proj } A})$

Prop. $\bullet \mathcal{F}_i \text{ qc} \Rightarrow \bigoplus_i \mathcal{F}_i \text{ qc}$
 $\bullet \mathcal{F} \text{ qc} (c) \Rightarrow f^* \mathcal{F} \text{ qc} (c)$

$\bullet f \text{ q-c, q-s } \mathcal{F} \text{ qc} \Rightarrow f_* \mathcal{F} \text{ qc}$
 $\bullet f \text{ proper } \mathcal{F} \text{ c} \Rightarrow f_* \mathcal{F} \text{ c}$

Prop. $\mathcal{G}$ locally free fin. rank, then $f_*(\mathcal{F} \otimes_{\mathcal{O}_X} f^* \mathcal{G}) \cong f_* \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}$

Prop. $\text{Hom}_A M, \Gamma(X, \mathcal{F}) \cong \text{Hom}_{\mathcal{O}_X}(\tilde{M}, \mathcal{F})$

Def. Support of q -c $\text{supp } \mathcal{F} = V(\text{Ann } M)$ closed

Prop. $\bullet F \text{ c, } \mathcal{F}_x \text{ free} \Rightarrow \mathcal{F}|_U \text{ free for some } U \ni x$
 $\bullet \mathcal{F} \text{ loc. free} \Leftrightarrow \mathcal{F}_x \text{ free } \forall x$

Prop. A Noeth. $+ M$ f.g. $\Rightarrow \tilde{M}$ coherent

Def (n -th twist). $\mathcal{F}(n) = \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_X(n)$

Prop. $\mathcal{O}_X(n)$ invertible, $\mathcal{O}_X(n+m) \cong \mathcal{O}_X(n) \otimes \mathcal{O}_X(m)$

Def (Extension). X Noetherian, $\mathcal{F}$ coherent on U , then $\exists \mathcal{G}$ coherent on X st $\mathcal{G}|_U \cong \mathcal{F}$.

Invertible, Picard and Divisors

$f : X \rightarrow Y, \mathcal{L}$ invertible.

Def. $\bullet$ (quasi-projective) $\exists$ immersion $\iota : X \rightarrow \mathbb{P}_Y^n$ over Y .
 $\bullet$ (projective) quasi-projective + proper.
 $\bullet$ (very ample) $\mathcal{L} \cong \iota^* \mathcal{O}_{\mathbb{P}_Y^n}(1), \iota : X \rightarrow \mathbb{P}_Y^n$ closed immersion over Y .
 $\bullet$ (ample) $\forall \mathcal{F} \in \text{Coh}(X) \exists n_0 > 0$ s.t. $\mathcal{F} \otimes \mathcal{L}^{\otimes n}$ g.g. for $n \geq n_0$

Prop. very ample $\Rightarrow$ ample

Prop. $\bullet \mathcal{L}, \mathcal{M}$ ample $\Rightarrow \mathcal{L} \otimes \mathcal{M}$ ample
 $\bullet \mathcal{L}$ ample, $\mathcal{M}$ g.g. $\Rightarrow \mathcal{L} \otimes \mathcal{M}$ ample
 $\bullet \mathcal{L}$ ample $\mathcal{L}^{\otimes n} \otimes \mathcal{M} \forall n \geq n_0$
 $\bullet \mathcal{L}$ very ample, $\mathcal{M}$ g.g. $\Rightarrow \mathcal{L} \otimes \mathcal{M}$ very ample

Prop. (X Noetherian) $\mathcal{L}$ ample $\Leftrightarrow \mathcal{L}^{\otimes n}$ ample $\forall n > 0 \Leftrightarrow \exists n > 0, \mathcal{L}^{\otimes n}$ ample

Prop. $f : X \rightarrow \text{Spec } A$ finite type, A Noetherian. Then $\mathcal{L}$ ample $\Leftrightarrow \exists n > 0, \mathcal{L}^{\otimes n}$ very ample.

Prop. $f : X \rightarrow \text{Spec } A$ proper, A Noetherian. Then $\mathcal{L}$ very ample $\Rightarrow \mathcal{L}$ ample.

Def. $\text{Pic}(X)$ group of iso. classes of inv. sheaves under $\otimes$.

Def (Dimensions). $\bullet \dim(X) := \sup\{n : Z_0 \subsetneq Z_1 \subsetneq \dots \subsetneq Z_n \subseteq X\}$ Z_i irred. closed.
 $\bullet \text{codim}(Z, X) := \sup\{n : Z = Z_0 \subsetneq \dots \subsetneq Z_n \subseteq X\}$ Z_i irred. closed.
 $\bullet \text{ht}(\mathfrak{p}) := \sup\{n : \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \dots \subsetneq \mathfrak{p}_n = \mathfrak{p}\}$

Prop. $\text{ht}(\mathfrak{p}) = \text{codim}(V(\mathfrak{p}), \text{Spec } A), \dim(\text{Spec } A) = \dim(A)$

Prop. A integral f.t. over k . Then $\dim A = \text{tr.deg}_k(\text{Frac}(A)) = \dim(A/\mathfrak{p}) + \text{ht}(\mathfrak{p})$

Prop (Krull's Hauptidealsatz). A Noetherian, a nonzerodivisor. Then $\mathfrak{p} \in V(a)$ with $\mathfrak{p}$ minimal $\Rightarrow \text{ht}(\mathfrak{p}) = 1$.

Prop. *A Noetherian, integral. Then $A = \bigcap_{\text{ht}(\mathfrak{p})=1} A_{\mathfrak{p}}$.*

Def (Regular). *Noetherian + local + $\dim_k \mathfrak{m}/\mathfrak{m}^2 = \dim A$.*

Prop. *(A Noetherian local dim 1) Then, $A \text{ regular} \Leftrightarrow A \text{ normal} \Leftrightarrow A \text{ dvr}$*

Def. • $\text{div} : \{(\mathcal{L}, s)\} / \sim \rightarrow \text{Div}(X), (\mathcal{L}, s) \mapsto \sum \nu_Z(s)Z$

- (effective) $n_Z \geq 0 \forall Z$
- (principal) $D = \text{div}(f)$ for some $f \in K(X)^*$
- (cartier) D closed subscheme whose ideal sheaf is invertible

Def (R1). *Regular in codim 1 $\Leftrightarrow \mathcal{O}_{X,x}$ regular $\forall x$ with $\dim(\mathcal{O}_{X,x}) = 1$.*

Prop. *Normal + Noetherian $\Rightarrow$ (R1)*

Prop. $\text{Pic}(X) \rightarrow \text{Cl}(X), [\mathcal{L}] \mapsto [\text{div}(s)]$

Prop. *X integral Noetherian (R1) $\Rightarrow \text{Pic}(X) \hookrightarrow \text{Cl}(X)$ injective.*

Def (Sheaf associated). $\mathcal{O}_X(D) : U \mapsto \{f \in k(X)^\times : \text{div}|_U f + D|_U \geq 0\} \cup \{0\}$

Prop. *Integral + normal + Noetherian $\Rightarrow \mathcal{O}_X(D) \in \text{QCoh}(X)$.*

Prop. $\mathcal{O}_X(\text{div}(\mathcal{L}, s)) \cong \mathcal{L}, 1 \mapsto s. D \text{ locally principal} \Rightarrow \mathcal{O}_X(D) \text{ invertible.}$

Prop. $\mathcal{O}_X(D) \text{ invertible} \Rightarrow D = \text{div}(\mathcal{O}_X(D), 1)$

Prop. • *A Noetherian + integral + $(\text{ht}(\mathfrak{p}) = 1 \Rightarrow \mathfrak{p} \text{ princ.}) \Rightarrow A \text{ factorial.}$*

- *A factorial $\Rightarrow A \text{ normal.}$*
- *A regular + Noetherian + local $\Rightarrow A \text{ factorial.}$*

Def. *A local integral, $k = \text{frac}(A)$.*

- $B \subseteq k$ local integral dominates $A \Leftrightarrow \mathfrak{m}_B \cap A = \mathfrak{m}_A$
- (vr) maximal with respect to domination.
- (dvr) vr + Noetherian + not field

Prop. *A dvr $\Leftrightarrow A$ local integral + $\mathfrak{m}$ principal*

Prop. *Valuative criterion.*

- $f \text{ q-c} \Rightarrow (f \text{ univ closed} \Leftrightarrow \text{existence})$
- $f \text{ q-s} \Rightarrow (f \text{ separated} \Leftrightarrow \text{uniqueness})$

$$\begin{array}{ccc} \text{Spec}(K) & \longrightarrow & X \\ \downarrow & \nearrow & \downarrow f \\ \text{Spec}(R) & \longrightarrow & Y \end{array}$$

Def. *X factorial $\Leftrightarrow \mathcal{O}_{X,x}$ factorial.*

Prop. *X factorial + integral + Noetherian $\Rightarrow \text{Pic}(X) \cong \text{Cl}(X)$*

Prop. *A factorial $\Rightarrow \text{Pic}(\text{Spec } A) = 0$*

Prop. • $\text{Cl}(X) \twoheadrightarrow \text{Cl}(U)$ surjective

- $\text{codim}(Z, X) \geq 2 \Rightarrow \text{Cl}(X) \cong \text{Cl}(X - Z).$
- $\text{codim}(Z, X) = 1 \Rightarrow \mathbb{Z} \rightarrow \text{Cl}(X) \rightarrow \text{Cl}(X - Z) \rightarrow 0$ exact.

Prop. $\text{Pic}(\mathbb{A}_k^n) = 0, \text{Pic}(\mathbb{P}_k^n) \cong \mathbb{Z} \cdot \mathcal{O}_{\mathbb{P}_k^n}(1)$

$$\begin{array}{ccccc} \{(\mathcal{L}, s)\} / \sim & \xrightarrow{\text{div}} & \{D \in \text{Div}(X) : D \text{ loc. princ.}\} & \longrightarrow & \text{Div}(X) \\ \downarrow & & \downarrow & & \downarrow \\ \text{Pic}(X) & \longrightarrow & \{D \in \text{Div}(X) : D \text{ loc. princ.}\} & \longrightarrow & \text{Cl}(X) \end{array}$$

Differentials

A ring, $B \in A\text{-algebra}, M \in B\text{-mod}, f : X \rightarrow Y$

Def. *Derivation $d : B \rightarrow M, A\text{-linear}$*

- $d(b + b') = d(b) + d(b')$
- $d(bb') = bd(b') + b'd(b)$
- $d(a) = 0 \forall a \in A$

Def. *Module of relative diff. forms $\Omega_{B/A}, d : B \rightarrow \Omega_{B/A}$ st*

$$\begin{array}{ccc} B & \xrightarrow{d'} & M \\ d \downarrow & \nearrow \exists! & \\ \Omega_{B/A} & & \end{array}$$

Prop. $B = A[x_1, \dots, x_n] \Rightarrow \Omega_{B/A} \cong \bigoplus_A dx_i$

Prop. • $\Omega_{B \otimes_A C/A} \cong (\Omega_{B/A} \otimes_A C) \oplus (\Omega_{C/A} \otimes_A B)$

- $\Omega_{S^{-1}B/A} \cong S^{-1}\Omega_{B/A}$

Def. *Cotangent sheaf $\Omega_{X/Y}$*

Prop. $g : Z \rightarrow Y$

- $\Omega_{X \times_Y Z/Y} \cong p^* \Omega_{X/Y} p^* \Omega_{Z/Y}$
- $\Omega_{U/Y} \cong \Omega_{X/Y}|_U$
- $\Omega_{X/V} \cong \Omega_{X/Y}$

Prop. *ses, $g : Y \rightarrow Z, Z \subseteq X$ closed ideal sheaf I*

- (cotangent) $f^* \Omega_{Y/Z} \rightarrow \Omega_{X/Z} \rightarrow \Omega_{X/Y} \rightarrow 0$
- (conormal) $I/I^2 \rightarrow \Omega_{X/Y} \otimes_{\mathcal{O}_X} \mathcal{O}_Z \rightarrow \Omega_{Z/Y} \rightarrow 0$